

Statistics 250, Section 701, Quiz 1 (75 Points)

May 28, 1998, Dr. Jürgen Symanzik

Your Name: _____

Question 1: Numbers and Graphs (40 Points)

The following questions are based on the data for the “Weider Empire” from USA Today. **When you write down your answers, indicate which values are assigned to the variables you use (e.g., $n = 20$, $x_{(15)} = 20,000$) and which formula you use.**

1. Determine the mean magazine circulation for the “Weider” empire. (6 Points)
2. Determine the median magazine circulation for the “Weider” empire. (5 Points)
3. Assume you are the press speaker of the “Smith” magazine group that also has 10 magazines with a mean circulation of 310,000 and a median circulation of 300,000. Which number(s) would you report when comparing your group with the “Weider” empire. Explain. (5 Points)
4. Calculate the range of the magazine circulation for the “Weider” empire. (3 Points)

5. Assume the “Weider” group possesses only these 10 magazines. Calculate the variance of the magazine circulation. Make a clear statement whether you are calculating a population or a sample variance. **(8 Points)**

6. Construct a stem-and-leaf display of the magazine circulation. **(6 Points)**

7. Are you happy with the data provided by USA Today? Isn't some important information missing? **(3 Points)**

8. Do you think that the 1.1 million circulation for “Shape” is the absolute truth? Think of possible manipulations (hint: “TV Guide” published 4 collector's covers of Seinfeld in their May 9–15 issue — how might this effect the circulation?). **(4 Points)**

Question 2: Frequencies and Histograms (20 Points)

Look at the blood donors data from Exercises 2.2, 2.35 & 2.37 in Blaisdell again. The ages of 40 donors from a computer company in Fairfax, Virginia, that visited a bloodmobile unit at their company are given as:

1. Construct a frequency distribution of the ages. Use classes of equal width and a first class of 20–29. Also calculate the relative frequency (r.f.). **(6 Points)**
2. Construct a **relative frequency histogram** based on your frequency distribution from above. **(6 Points)**

3. Does this data seem to belong to a bell-shaped distribution? It might be useful to know that $\bar{x} = 39.1$, $\tilde{x} = 37.5$, and $s = 10.96$. (**4 Points**)

4. Based on this data, do you think it is correct to state that about 35% of all blood donors in the United States are in the 30 to 39 age group? Explain. (**4 Points**)

Question 3: Mosaic Plots (**15 Points**)

Look at the Mosaic Plot of the Titanic data again.

1. Mark the following regions, using the letters “A”, “B”, and “C”:
“A”: Second Class, Child, Male, Survived.
“B”: Third Class, Adult, Female, Not Survived.
“C”: Crew, Adult, Female, Survived.
(**6 Points**)
2. Indicate which of these statements are correct and which are incorrect:
(i) A very large percentage of First Class, Adult, Female survived.
(ii) Among the Adult, Male passengers, the least percentage that survived was in Second Class.
(iii) There were more Male than Female children on board of the Titanic.
(**9 Points**)